Parameter	Symbol	Domain	Definition
Planned start	s_i	$\mathbb{Z}_{\geq 0}$	Scheduled period at which project i begins.
Planned duration	D_i	$\mathbb{Z}_{\geq 1}$	Number of active periods planned for project i.
Planned finish	f_i	$\mathbb{Z}_{\geq 0}$	$f_i = s_i + D_i$; derived at reset.
S-curve shape	$a_i,\ b_i$	$\mathbb{R}^+$	Beta CDF shape parameters governing $PV_i(\tau)$.
Progress delay cap	$\delta_i^{\max}$	$[0,1]$	Maximum admissible progress lag before the breach flag fires.
Finish delay cap	$\Delta_i^{\max}$	$\mathbb{Z}_{>0}$	Extra periods beyond f_i before the hard deadline fires.
Termination tolerance	θ_i	$\mathbb{Z}_{>0}$	Grace-period length; models FIDIC Sub-Clause 15.1 (FIDIC, 2017).
Budget at completion	BAC_i	$\mathbb{R}_{>0}$	Total planned contractor cost.
Profit margin	π_i	$\mathbb{R}_{\geq 0}$	Contract price $P_i = \mathrm{BAC}_i(1+\pi_i)$.
Advance payment rate	ϕ_i	$[0, 0.3]$	Advance $\mathrm{Adv}_i = \phi_i P_i$ paid at $t = s_i$; 0 disables it.
Advance recovery threshold	ψ_i^{rec}	$[0,1]$	Certified gross fraction above which recovery deductions start.
Advance recovery rate	δ_i^{rec}	$[0,1]$	Per-milestone deduction rate used to recover Adv_i.
Retention rate	ν_i	$[0, 0.1]$	$R_{i,j}^{\mathrm{held}} = \nu_i G_{i,j}$; released in full at $j = M_i$.
Milestone weight	$w_{i,j}$	$[0,1]$	Fraction of P_i paid at milestone j; $\sum_j w_{i,j} = 1$.
Milestone progress threshold	$\psi_{i,j}$	$(0,1]$	Minimum $EV_{i,t}$ to certify milestone j; $\psi_{i,M_i} = 1$.
Milestone timestep threshold	$\tau_{i,j}^{\mathrm{ms}}$	$\mathbb{Z}_{\geq 0}$	Earliest project time at which milestone j can be certified.
Initial budget	B_0	$\mathbb{R}_{>0}$	$B_0 = \kappa \sum_i \mathrm{BAC}_i$; derived at reset.
Budget tightness	κ	$\mathbb{R}_{>0}$	Liquidity regime scalar.
Discount factor	γ	$(0,1]$	Per-period discount on net cash-flow rewards (Damodaran, 2012).
Annual interest rate	r	$[0,1]$	Monthly treasury-draw cost $= r/12$.
Episode horizon	T	$\mathbb{Z}_{>0}$	Derived quantity; not a configuration entry.
Work efficiency	$\eta_{i,t}$	≥ 0.01	Sole stochastic primitive; drawn per period per active project.
Portfolio size	N	$\mathbb{Z}_{>0}$	Number of projects; sampled at each reset if a distribution.
Milestone count	M_i	$\mathbb{Z}_{>0}$	Number of milestones of project i.